

Code Challenge

Objective: Develop a backend application to perform an ETL using AWS Services.

Functional Requirements:

1. AWS Services Integration:

- Use EventBridge to trigger the Lambda at a scheduled time.
- Use AWS Lambda to start the Step Function.
- Use Glue Jobs (Not notebook) to process data.
- Coordinate the Glue Job using a Step Function.
- Use S3 as the data and resource storage.
- Use Data Catalog for Iceberg Table creation.

Technical Requirements:

- Use AWS SAM (Serverless Application Model) to define and deploy the infrastructure declaratively.
- Write the Lambda functions and Glue Job in Python/PySpark.
- Document the architecture and deployment steps in a README.md file.

Detailed Application Flow:

Data description:

Use the [battery14_df.csv](#) and upload it to s3.

Create a new **degrees.csv** from this data and upload it to s3.

```
None
education_level,description
1,Some high school
2,High school diploma/GED
3,Some college
4,College degree
5,Professional degree
6,Master's degree
7,Ph.D.
8,Associate's degree
99,Other
```

Application stages:

1. ETL starts:

- A schedule triggers a Lambda function to start the State Machine. ○ Lambda can adjust the State Machine parameters (If required).

2. ETL orchestrator:

- State Machine coordinates the required Glue Jobs
- A Glue Job reads data from S3, the **battery14_df.csv**, and the **degrees csv**.
 - i. Select all females over 30 years old from the US who have a higher education level than a Master's degree and a raw score greater than 300.
 - ii. Apply some transformations:
 - 1. Replace **m** with Male and **f** with Female
 - 2. Join the results with the proper degree name.
 - iii. Write the resulting dataframe to a new S3 bucket.

3. Iceberg Table:

- A final glue job reads the transformed data and creates an Iceberg Table in Glue Data Catalog.

4. Validate Results:

- To validate the creation. The new table must be queried using SQL PySpark or AWS Athena.

Evaluation:

- Source code demonstrating effective and efficient use of AWS Lambda, Step Functions, Glue jobs, and PySpark.
- Documentation explaining the implemented architecture and instructions for testing the solution.